

ARCO

Arquitecture and Networks Research Group

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http://arco.esi.uclm.es

- **School of Computer Science** at the University of Castilla-La Mancha



CIUDAD REAL
(50 minutes away from Madrid in AVE)

Members

- Staff (Professors and Lecturers)
  - 8 Full & Associate
  - 3 PhD candidates
- Other staff
  - 4 Computer Science Engineers
  - 4 Graduate students

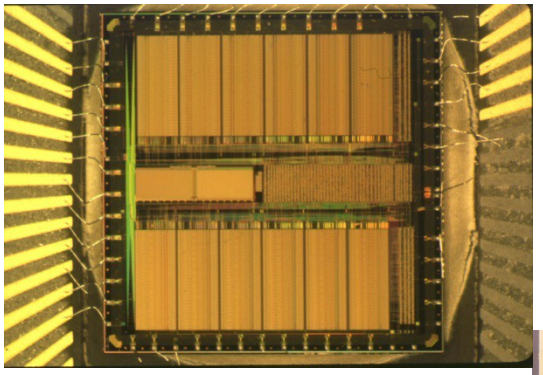
Research lines

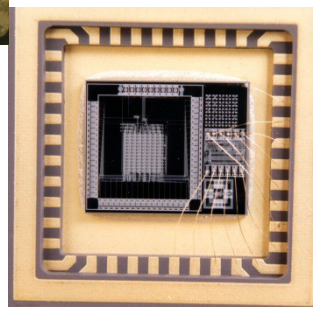
- Heterogeneous System Design
- Distributed systems
- Artificial intelligence: Common Sense Reasoning

Our "Customers" are developers!

Research topics (I)

- Hetereogeneous (Hardware-Software) System Design
  - HW-SW codesign
  - Design Methodologies
  - Embedded system design
  - Platform-based design

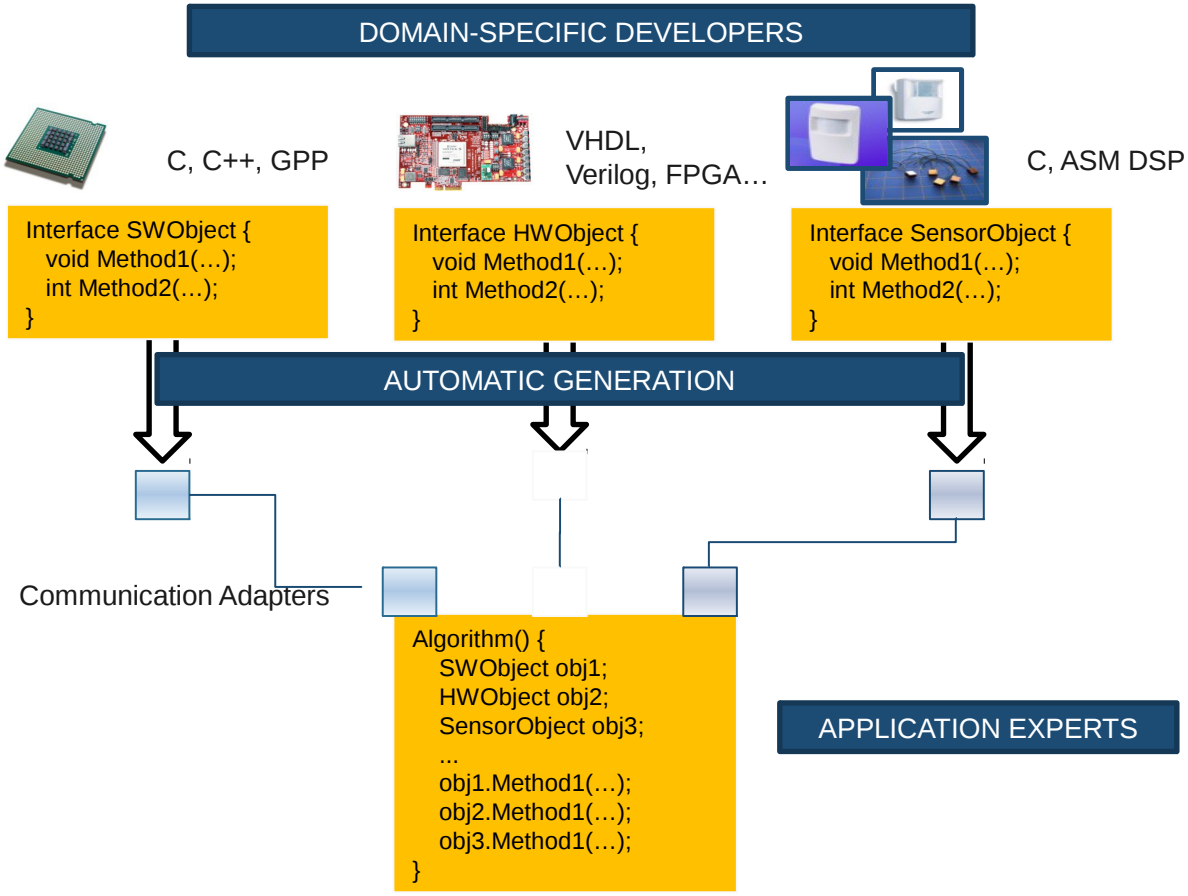




- Design of reconfigurable architectures (FPGAs)

Reconfigurable System Design

MHAL Standard implementation in FPGAs and DSPs using our middleware technology



The RGrid project → H2020

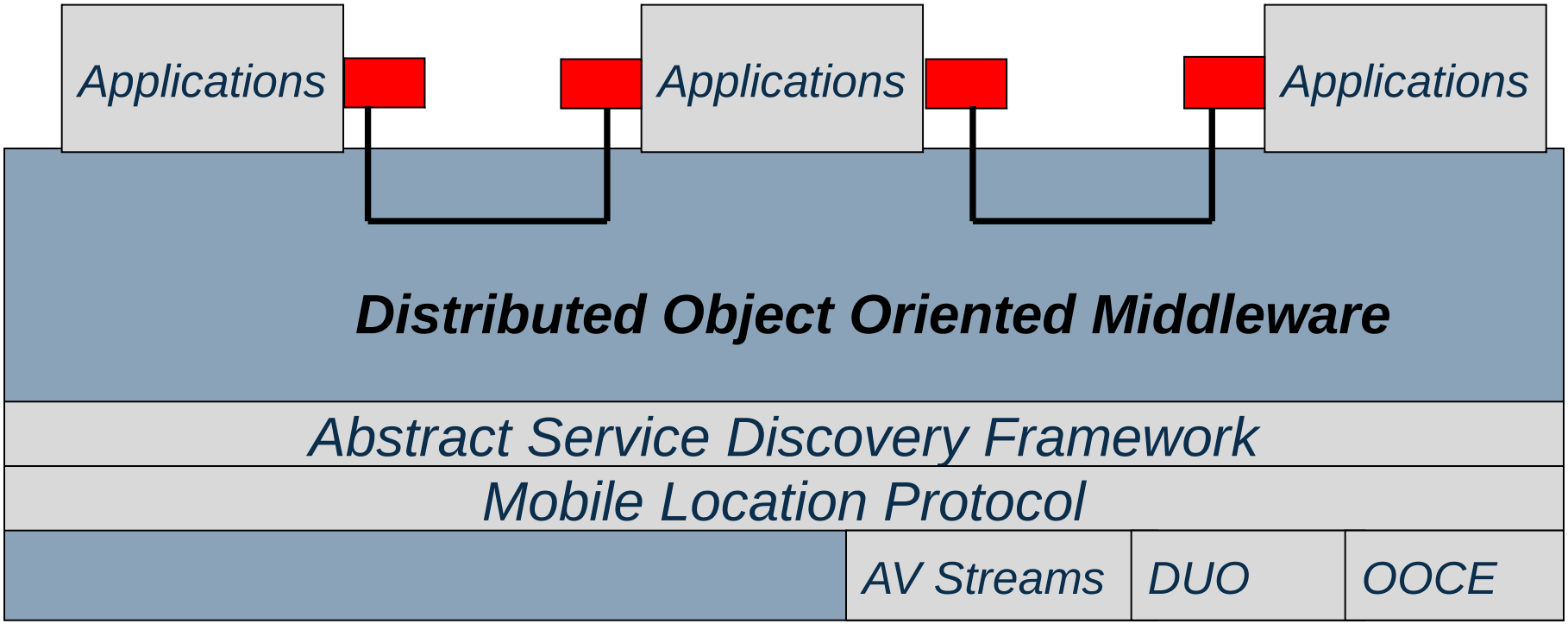
- The goal is to offer reconfigurable computing resources as traditional computing ones (e.g. CPU time) in Grid architectures.
  - Remote, decentralized and **transparent** access, management and deployment of functionality into FPGAs
  - Homogeneous programming environment for RGrid



Research topics (II)

- Distributed systems and communication infrastructures for
  - Smart Cities, Ambient Intelligence and Internet of Things.
- Mainly using the **object-oriented paradigm** we are:
  - **Integrating** low footprint devices, FPGAs, audio/video devices, etc.
  - **Developing tools** for generating the communication infrastructure in automatic way.
    - Removing error-prone manual tasks
  - **Enabling engineers** to use HW algorithms, sensors, cameras, etc. as distributed objects.
    - Reducing the learning curve

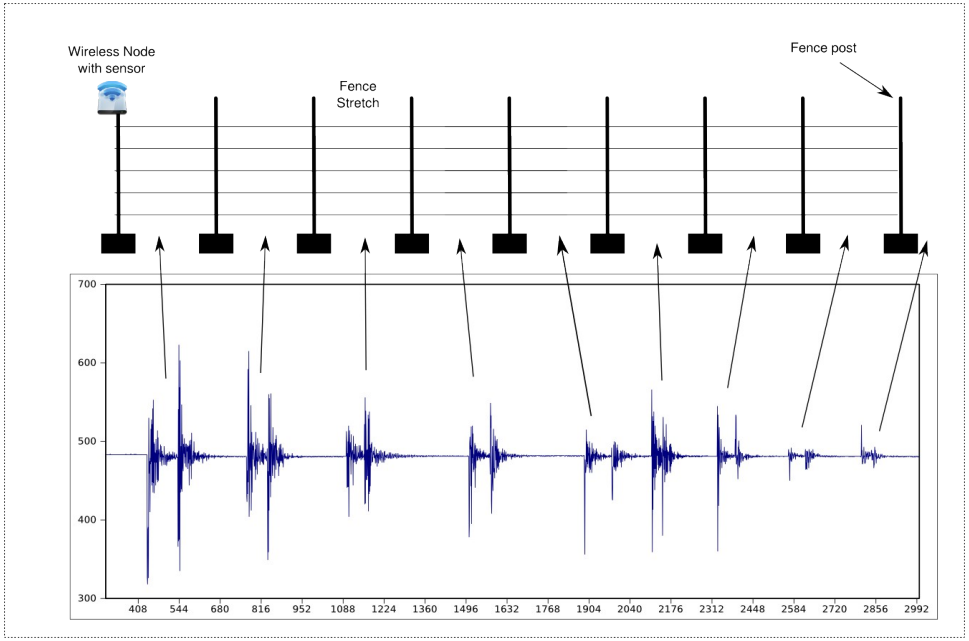
Integration of heterogeneous systems





VIA project: Fence monitoring in Highways







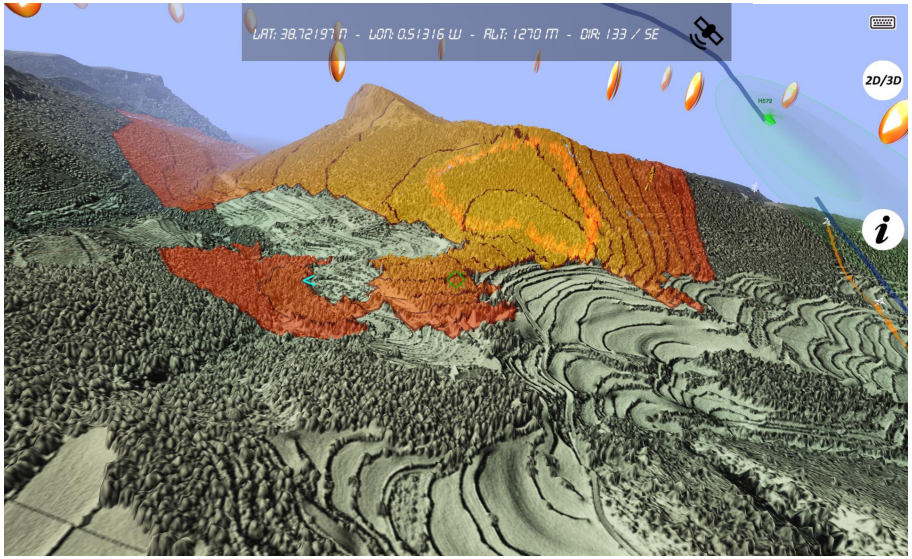
Previous works: HESPERIA

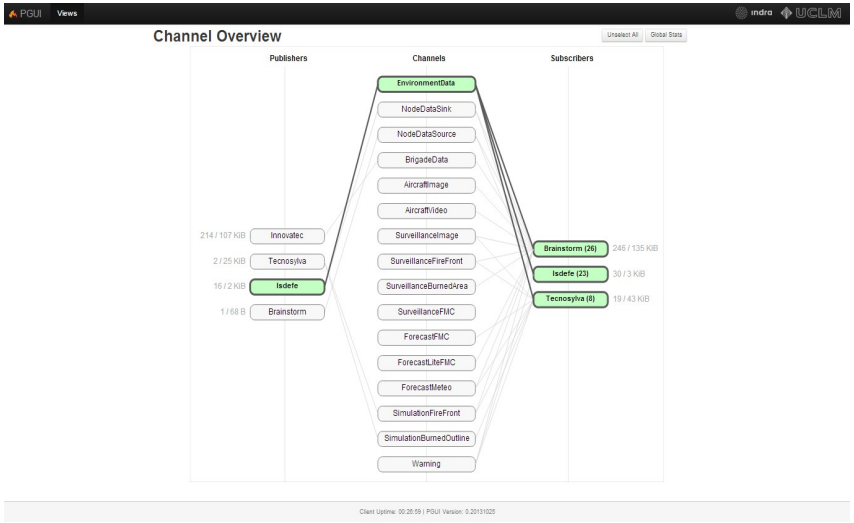
- Seamless integrated management for homeland security
- 7 Companies
- 9 Universities and Research institutes
- 23 Million Euros (4 years)

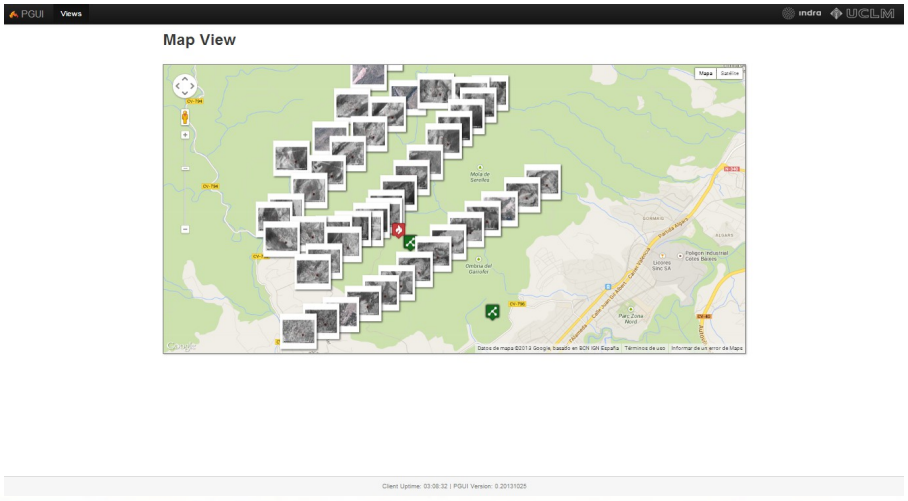


Promete

UCLM
UNIVERSIDAD DE CASTILLA-LA MANCHA



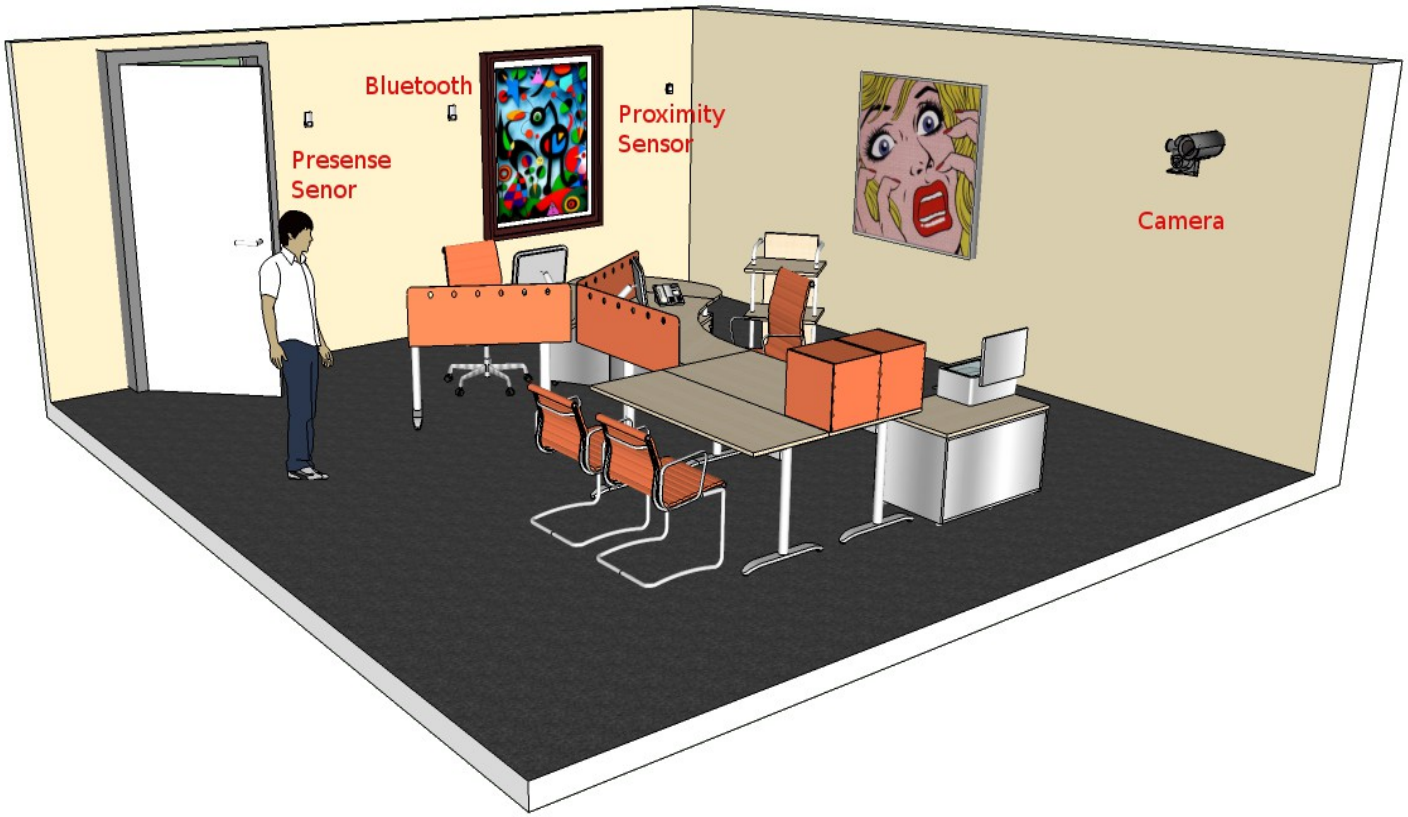




Research topics (and III)

- **Smart** City, Ambient **Intelligence** → **Smart Middleware**
- Bringing Intelligence into middleware
- Common Sense reasoning for:
  - Automatic service composition
  - Resource management
  - Situation identification
- To provide smart services to developers
- Using Scone, a common sense reasoning engine (Carnegie Mellon University)

Research topics (and III)



Motion detection = Presence Sensor
Motion detection = Camera + Algorithm

Projects

Prometeo

2010-2013

indra

energos

2009-2012

Software-Defined Radio Integration Platform

Hesperia

2006-2009

indra CÁTEDRAS > UNIVERSIDAD DE CASTILLA - LA MANCHA

ElCano

Multimodal Indoor Navigation Infrastructure for Disabled People

Projects

- DOMOBOX



- DREAMS, Seinveh, SAND, DAMA, RGRID, Consignos, etc.

- AGIL





- TETRA





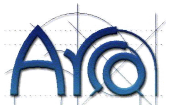
- DEMETER



InfoGLOBAL







eDIGIREGION

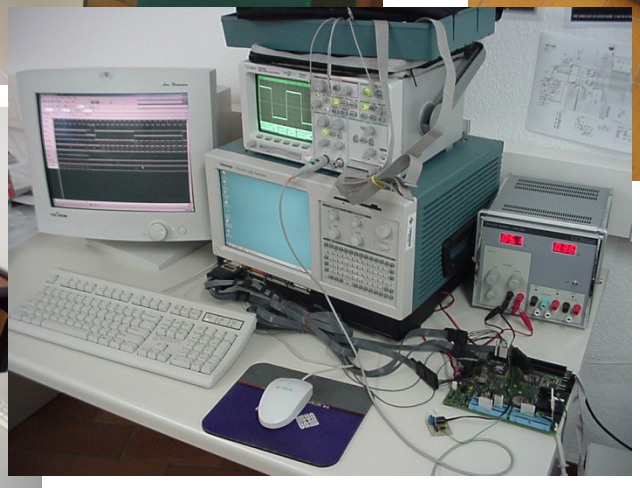
- FP7 Programme
- Thirty-six month
- Main Purspose: To increase regional competitiveness via research-driven clusters in the technology domain.
- Partners:
  - Southeast Ireland
  - Central Hungary
  - Bucharest-Ilfov, Romania
  - Castilla-La Mancha, Spain.

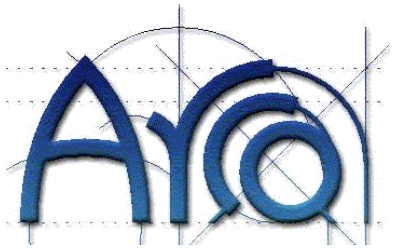
Our laboratory











More information at:

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